



## DSA MARATHON

# Pattern Printing Cheat Sheet

*Topic: Pattern Printing*

## 1. Introduction

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### What is Pattern Printing?

Pattern Printing is the process of printing different shapes (stars, numbers, alphabets, symbols) using loops.

It is one of the best ways to understand:

- Nested Loops
- Loop execution
- Logic building
- Problem-solving skills

### Why do we learn Pattern Printing?

- Improves logical thinking.
- Builds confidence in nested loops.
- Helps in coding interviews.
- Strengthens loop concepts.
- Prepares for DSA problems.

## 2. Basic Concepts

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### Outer Loop

The outer loop controls the number of rows.

```
*****  
*****  
*****
```

```
Rows = 3
```

```
C++
```

```
for (int i = 1; i <= 3; i++)
```

### Inner Loop

The inner loop controls the number of columns.

```
*****
```

```
Columns = 5
```

```
C++
```

```
for (int j = 1; j <= 5; j++)
```

## Relation between Outer and Inner Loop

```
Outer Loop
```

```
|
```

```
▼
```

```
Rows
```

```
|
```

```
▼
```

```
Inner Loop
```

```
|
```

```
▼
```

```
Columns
```

## 3. General Pattern Logic

Every pattern follows four steps.

### Step 1 — Count Rows

```
*
**
***
****
```

```
Rows = 4
```

### Step 2 — Count Columns

Each row has different columns.

### Step 3 — Find Relationship

```
Row 1 → 1 Star
Row 2 → 2 Stars
Row 3 → 3 Stars
Row 4 → 4 Stars
```

**Formula:** Stars = Row Number

### Step 4 — Write Loops

```
C++
```

```
for (row)
```

```
{  
    for (column)  
    {  
        print;  
    }  
}
```

## 4. Pattern Solving Approach

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Whenever you see a pattern, follow these steps.

```
Observe Pattern  
  |  
  ▼  
Count Rows  
  |  
  ▼  
Count Columns  
  |  
  ▼  
Find Formula  
  |  
  ▼  
Write Outer Loop  
  |  
  ▼  
Write Inner Loop  
  |  
  ▼  
Print
```

## 5. Types of Patterns

---

### Star Pattern

```
*****  
*****  
*****
```

### Right Triangle

```
*  
**  
***  
****
```

### Inverted Triangle

```
****  
***  
**  
*
```

### Left Triangle

```
  *  
 **  
***  
****
```

### Pyramid

```
  *  
 ***  
*****  
*****
```

### Inverted Pyramid

```
*****  
****  
***  
*
```

### Diamond

```

      *
    ***
  *****
*****
  *****
    ***
      *

```

## Hollow Square

```

*****
*      *
*      *
*      *
*****

```

## Hollow Pyramid

```

      *
    *  *
  *    *
*****

```

## Butterfly Pattern

```

*      *
**     **
***   ***
*****
***   ***
**     **
*      *

```

## 6. Number Patterns

---

Example

```
1
12
123
1234
```

### Reverse Number Pattern

```
1234
123
12
1
```

### Floyd's Triangle

```
1
2 3
4 5 6
7 8 9 10
```

### Pascal's Triangle

```
      1
     1 1
    1 2 1
   1 3 3 1
```

## 7. Alphabet Patterns

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```
A
AB
ABC
ABCD
```

### Reverse Alphabet

```
ABCD
ABC
AB
A
```

## 8. Spaces Logic

Most students struggle with spaces.

Example

```
*
***
*****
```

Spaces Formula:  $\text{Spaces} = n - \text{row}$

## 9. Stars Formula

Right Triangle  $\rightarrow \text{Stars} = \text{Row Number}$

Reverse Triangle  $\rightarrow \text{Stars} = n - \text{row} + 1$

Pyramid  $\rightarrow \text{Stars} = 2 \times \text{row} - 1$

## 10. Nested Loop Visualization

Outer Loop

|

▼

Row 1

|

▼

Inner Loop

|

▼

Print

|

▼

Next Row

## 11. Dry Run

Example —  $N = 4$

```
N = 4
Row = 1
  Column = 1 → Print *
Next Row
Row = 2
  Column = 1 → Print *
  Column = 2 → Print *
...
```

## 12. Common Formulas

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### Quick Formula Recap

| Pattern          | Formula                                              |
|------------------|------------------------------------------------------|
| Right Triangle   | Stars = Row                                          |
| Reverse Triangle | Stars = $n - \text{row} + 1$                         |
| Pyramid          | Stars = $2 \times \text{row} - 1$                    |
| Spaces           | Spaces = $n - \text{row}$                            |
| Diamond          | Upper Half (Pyramid) + Lower Half (Inverted Pyramid) |

**End of Pattern Printing Cheat Sheet**

**NextGen Data Minds**